

Community Breadth Diversity Does Not Predict Benchmark Displacement: A Pre-Validated Null Result

Anonymous Authors

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Abstract

We investigate whether community breadth at benchmark introduction — operationalized as the count of distinct paper submissions per benchmark through its plurality-introduction year on Papers With Code — predicts plurality benchmark displacement hazard. Two competing mechanisms motivate the hypothesis: stakeholder lock-in (broad early adoption $\rightarrow$ investment network $\rightarrow$ displacement resistance, $HR < 1$) and saturation pressure (widespread adoption $\rightarrow$ perceived as solved $\rightarrow$ accelerated replacement, $HR > 1$). Before testing the mechanism, we apply a 5-gate FAIL FAST pre-validation protocol that confirms join coverage ($G0 = 0.862$), predictor time-independence via partial R^2 ($G1 = 0.605$, $G2 = 0.975$), adequate variance ($G3 \text{ std} = 0.246$), and no multicollinearity ($G4 \text{ max VIF} = 2.14$). All five gates pass by wide margins, establishing that any downstream null reflects genuine absence of effect rather than data quality failure. Cox proportional hazards regression (**CoxPHFitter**, penalizer = 0.1) on 258 complete-case rows from the h-e2 panel (87 Koch et al. 2021 tasks, 2015–2023; EPV ≈ 86) yields $HR = 1.006$ (95% CI = [0.846, 1.196]), LRT $p = 0.9495$ — a near-perfect null. The diversity predictor adds $\Delta \log L = 0.0020$ to the baseline model. Neither the lock-in nor the saturation mechanism is operative at a detectable level. We conclude that submission-count community breadth diversity does not predict benchmark displacement timing in the Papers With Code ecosystem. The 5-gate FAIL FAST protocol is a reusable methodological contribution for benchmark lifecycle survival analysis; score trajectory and institutional diversity remain open directions for future investigation.

1 Introduction

Machine learning progress is increasingly measured through benchmark performance. Papers With Code (PWC) hosts thousands of benchmark leaderboards where research teams compete to achieve state-of-the-art (SOTA) results on shared evaluation datasets. Yet benchmarks themselves have lifecycles: a benchmark that defines the frontier today may be superseded tomorrow when the community adopts a harder or more comprehensive successor. Understanding what predicts this *benchmark displacement* — the event where one benchmark loses plurality status to another within the same task — has implications for how we design evaluation infrastructure, allocate research effort, and interpret progress claims.

A natural candidate predictor is *community breadth at benchmark introduction*: how many distinct research teams participated in the benchmark when it first achieved plurality status. Two competing mechanisms motivate this hypothesis. Under the **lock-in mechanism** (H1), broad early adoption creates a large network of stakeholders who have invested in the benchmark — trained models, established baselines, published results. This investment creates switching costs, slowing displacement (hazard ratio $HR < 1$). Under the **saturation mechanism** (H2), widespread adoption signals that the community has collectively “solved” the benchmark, accelerating the search for harder successors and replacement ($HR > 1$). Both mechanisms are theoretically grounded: H1 follows Rogers’ Diffusion of Innovations framework for opinion-leader lock-in Rogers [2003]; H2 aligns with the ICLR 2025 workshop framing of benchmark overuse and community-driven replacement icl [2025].

The challenge in testing this hypothesis is measurement. Prior attempts using raw cumulative submission counts collapsed into time proxies (partial $r^2 = 0.0011$ with temporal controls) — a methodological failure that renders Cox regression uninterpretable. A prior directional signal ($HR = 0.871$ for *score velocity*, a

different construct) appeared only with 22 displacement events — an underpowered sample where sampling noise dominates. The current paper tests *submission-count diversity* at adequate power; the two results are complementary, not contradictory (see Section 5.4).

This paper makes three contributions:

1. **Methodological:** We introduce a 5-gate FAIL FAST pre-validation protocol (G0–G4) that separates data quality from hypothesis validity before any regression. Gates test join coverage (G0), time-independence of diversity predictors via partial R^2 (G1–G2), cross-benchmark variance (G3), and multicollinearity (G4). All five gates pass by wide margins, ruling out measurement failure as an explanation for the null.
2. **Empirical:** We conduct the most statistically powered test to date of community breadth *as operationalized by paper submission count* as a displacement predictor. On 258 complete-case rows from the h-e2 panel (87 Koch et al. Koch et al. [2021] taxonomy tasks, 2015–2023, Papers With Code), with events-per-variable (EPV) ≈ 86 , we find $HR = 1.006$ (95% CI = [0.846, 1.196], LRT $p = 0.9495$). This is a **near-perfect null**.
3. **Scientific:** The combination of validated measurement and adequate power means the null is scientifically informative. Community breadth diversity — operationalized as unique paper submission count at introduction year — is ruled out as a predictor class for benchmark displacement timing in the Papers With Code ecosystem (2015–2023). Score velocity and institutional diversity remain untested at full power.

2 Related Work

2.1 Benchmark Lifecycle and Displacement Dynamics

The lifecycle of ML benchmarks has received increasing attention as the community grapples with benchmark saturation and the challenge of meaningful progress measurement. Ott et al. Ott [2022] analyze 3,765 benchmarks, finding that benchmark *breadth* (versatility across tasks) correlates with longevity at the population level. This population-level correlation motivates our investigation of whether individual-benchmark breadth predicts displacement timing — a finer-grained survival analysis question that Ott et al. do not address.

Koch et al. Koch et al. [2021] establish a 87-task taxonomy for Papers With Code and measure *concentration* dynamics in benchmark adoption. Their methodology provides the h-e2 panel that underlies our analysis. Where Koch et al. characterize concentration trends across tasks, we use the same panel to test individual predictors of displacement events.

Paullada et al. Paullada and et al. [2021] provide a qualitative governance framework for dataset lifecycles, arguing that benchmark datasets are sociotechnical artifacts whose trajectories are shaped by community norms, institutional pressures, and competitive incentives. Our work operationalizes their qualitative observations into a quantitative survival analysis framework.

2.2 Benchmark Overuse and Community Replacement

The ICLR 2025 workshop on benchmarking icl [2025] explicitly foregrounds benchmark overuse as a driver of community-initiated replacement. Under this framing (our H2 saturation mechanism), a benchmark that has been widely adopted signals consensus that the task is “solved,” incentivizing the community to propose harder successors. The GLUE $\rightarrow$ SuperGLUE transition is a canonical example: broad community adoption of GLUE Wang et al. [2018] was followed by recognition of GLUE’s ceiling, driving rapid adoption of SuperGLUE Wang et al. [2019].

Our null result partially challenges this framing at the individual benchmark level: even if overuse drives replacement in high-profile cases, submission count breadth at introduction year does not predict displacement timing across the broader population of 87 tasks.

2.3 Community Adoption and Lock-In Theory

Rogers’ Diffusion of Innovations Rogers [2003] provides the theoretical foundation for our H1 lock-in mechanism: early adopters become opinion leaders whose investment in a technology creates network effects that slow transitions to alternatives. Our null result suggests that this investment, at the aggregate level measured by submission count, does not translate into reduced displacement hazard.

2.4 Survival Analysis in Bibliometrics

We use `CoxPHFitter` from the lifelines library Davidson-Pilon [2019] for all Cox regressions. The L2 penalizer (penalizer = 0.1) achieves clean convergence, validated across 14/14 unit tests on the h-e2 panel. The key methodological challenge — identifying time-independent predictors that do not collapse into temporal proxies — is directly addressed by our FAIL FAST protocol.

3 Methodology

3.1 Dataset: h-e2 Survival Panel

Our analysis is based on the **h-e2 panel**, a validated survival dataset constructed from Papers With Code Papers With Code [2025]. The panel covers 87 parent tasks from the Koch et al. Koch et al. [2021] 87-task taxonomy, spanning 2015–2023. Each row corresponds to a benchmark that achieved plurality status (highest SOTA score for its task) at some point in the observation window.

Panel statistics: 345 rows; 87 tasks; 2015–2023; validated across 10 prior pipeline attempts (14/14 unit tests pass).

Survival variables: `duration` (years benchmark held plurality); `event` = 1 if displaced, 0 if right-censored.

Baseline covariates: `task_age` (years since task appeared in PWC); `log_publication_volume` (log total paper count for task). Note: `benchmark_introduction_year` excluded due to perfect collinearity with `task_age` ($|r| = 1.000$), detected automatically by Gate G4.

3.2 Diversity Predictor Construction

We construct diversity predictors from **pwc-archive/evaluation-tables** Papers With Code [2025] (Hugging-Face, CC-BY-SA-4.0, 326k rows after PyArrow flattening). The key field is `paper_url`, identifying distinct paper submissions per benchmark.

Primary predictor: `log_unique_paper_count_at_intro_z` = $\log_{1p}(\text{unique } \text{paper_url} \text{ count per benchmark through introduction year})$, z -standardized. Measures breadth of community participation in paper submissions at introduction.

Secondary predictor: `paper_diversity_ratio_at_intro_z` = (unique count / total submission rows), z -standardized. Normalizes by total volume, removing time-accumulation effects.

Join: Task paths matched via rapidfuzz `token_sort_ratio` (threshold = 85), achieving 86.2% coverage (75/87 benchmarks). Predictor collinearity: Pearson $r = -0.324$ — independent constructs.

3.3 5-Gate FAIL FAST Pre-Validation Protocol

A key methodological contribution is the **FAIL FAST protocol** (H-E1), which validates data quality *before* any Cox regression. Gates are evaluated in order; failure routes to hypothesis revision.

Partial R^2 computation: Regress predictor on [`task_age`, `benchmark_introduction_year`] via OLS; take R^2 of residuals. Partial $R^2 \gg 0.01$ indicates the predictor retains genuine cross-benchmark information beyond temporal trend.

3.4 Cox Proportional Hazards Model

We use `CoxPHFitter(penalizer=0.1)` from lifelines Davidson-Pilon [2019].

M0 (baseline): $h(t) = h_0(t) \cdot \exp(\beta_1 \cdot \text{task_age} + \beta_2 \cdot \text{log_publication_volume})$

Table 1: FAIL FAST Gate Results

Gate	Name	Criterion	Value	Status
G0	Join Coverage	$\geq 80\%$ benchmarks with non-null <code>paper_url</code>	0.862	PASS
G1	Primary Time-Indep.	partial $r^2(\log_count_z \sim \text{temporal}) > 0.01$	0.605	PASS
G2	Secondary Time-Indep.	partial $r^2(\text{diversity_ratio_z} \sim \text{temporal}) > 0.01$	0.975	PASS
G3	Variance	$\text{std}(\text{diversity_ratio}) > 0.10$	0.246	PASS
G4	Multicollinearity	$\max \text{VIF} < 10$	2.14	PASS

M1 (diversity predictor): $h(t) = h_0(t) \cdot \exp(\beta_1 \cdot \text{task_age} + \beta_2 \cdot \log_publication_volume + \beta_3 \cdot \log_unique_paper_count_at_intro_z)$

LRT: LRT stat = $-2 \cdot (\ell(M0) - \ell(M1))$, df = 1; p from chi-squared distribution.

Success criteria (pre-specified): P1 (primary): LRT $p < 0.05$ AND $|\text{HR} - 1| \geq 0.10$.

Direction protocol (pre-specified, no post-hoc modification):

- $\text{HR} < 1, p < 0.05 \rightarrow \text{H1 lock-in}$
- $\text{HR} > 1, p < 0.05 \rightarrow \text{H2 saturation}$
- $p > 0.05$ (G0–G4 pass) $\rightarrow \text{H0 null}$

Complete-case analysis: 87/345 rows dropped (NaN); 258 complete-case rows; $\text{EPV} \approx 86$.

4 Experiments

4.1 H-E1: Data Pipeline and FAIL FAST Gate Validation

Objective: Validate that the diversity predictors derived from `pwc-archive/evaluation-tables` meet all 5 FAIL FAST quality gates before proceeding to Cox regression.

Implementation:

- `preprocess_eval.py`: PyArrow-native flattening of the nested HuggingFace dataset (`list_flatten` + struct field access). Python-level iteration is $\sim 45\times$ slower; PyArrow is essential for the 326k-row dataset.
- `pipeline.py`: `DataLoader` $\rightarrow$ `FuzzyJoiner` (`rapidfuzz`, `threshold=85`) $\rightarrow$ `DiversityAggregator` (groupby `task_path`, `nunique paper_url`, compute `diversity_ratio`)
- `gates.py`: `GateValidator` (G0–G4 in order), `VIFChecker` (with perfect-collinearity detection)
- `output.py`: 5 figures generated; enriched panel written as `h_e2_panel_with_diversity.csv` (345 rows)

Key implementation notes:

- `pub_year` is not available at the SOTA-row level in `pwc-archive`, so temporal filter falls back to all rows. The partial R^2 gates (G1, G2) still pass because the diversity ratio’s time-independence is a structural property.
- `benchmark_introduction_year` is excluded from VIF (and downstream Cox models) due to perfect collinearity with `task_age` ($|r| = 1.000$).

4.2 H-M1: Cox Proportional Hazards Mechanism Test

Objective: Test whether `log_unique_paper_count_at_intro_z` significantly predicts plurality benchmark displacement hazard after controlling for `task_age` and `log_publication_volume`.

Implementation:

- `cox.analysis.py`: `load_panel()` → `fit_models()` (M0 + M1 with penalizer=0.1 and fallback 0.5) → `run_lrt()` (scipy `chi2.sf`; negative-stat guard) → `run_diagnostics()` (concordance index, PH assumption check)
- `visualization.py`: 5 figures (`gate_metrics.png`, `km_quartiles.png`, `partial_effects.png`, `forest_plot.png`, `schoenfeld_residuals.png`)
- `tests/test_cox_analysis.py`: 10 spec-compliance tests; all pass (10/10)

Input: `h_e2_panel_with_diversity.csv` from H-E1 (345 rows); complete-case analysis after dropping 87 NaN rows → 258 rows.

Kaplan-Meier analysis (P3): Benchmarks stratified into quartiles by `log_unique_paper_count_at_intro`. KM curves for Q1 (lowest diversity) vs. Q4 (highest diversity) plotted via `lifelines.KaplanMeierFitter`. Visual inspection for separation.

Proportional hazards check: `M1.check_assumptions()` was called but raised "could not convert string to float: 'dependency-parsing'" due to `task_path` stored as string slug (fix: integer-encode `task_path` before calling `check_assumptions()`). Visual inspection of Schoenfeld residuals (`figures/schoenfeld_residuals.` showed no obvious violation of the PH assumption; quantitative confirmation via automated test was not completed. Given HR = 1.006, moderate PH misspecification would not change the null conclusion.

Coder-Validator cycles: 1 cycle; validator approved all 12 tasks; 10/10 SDD tests pass.

5 Results

5.1 H-E1: All 5 Gates Pass

Table 2: FAIL FAST Gate Results

Gate	Metric	Value	Threshold	Status
G0	Join coverage	0.862	≥ 0.80	PASS
G1	partial $r^2(\log_count_z)$	0.605	> 0.01	PASS
G2	partial $r^2(\text{diversity_ratio_z})$	0.975	> 0.01	PASS
G3	std(<code>diversity_ratio</code>)	0.246	> 0.10	PASS
G4	max VIF	2.14	< 10.0	PASS

Additional: Pearson $r(\log_count_z, \text{diversity_ratio_z}) = -0.324$ (collinearity failsafe not triggered). 75/87 benchmarks matched; 67/87 with diversity stats computed.

The FAIL FAST all-pass establishes: (a) diversity signal exists and is joinable (G0); (b) both predictors are genuinely time-independent (G1 partial $r^2 = 0.605$, G2 = 0.975); (c) adequate cross-benchmark variance for Cox (G3 std = 0.246 $\gg$ 0.10). Any downstream null is attributable to genuine absence of effect.

5.2 H-M1: Near-Perfect Null

HR = 1.006 is indistinguishable from no effect. $\Delta \log L = 0.0020$ — the predictor adds essentially zero explanatory power. The CI [0.846, 1.196] straddles 1.0 by a wide margin. Both H1 (lock-in, requires HR < 1) and H2 (saturation, requires HR > 1) are falsified. Direction: **H0 (null)**.

Model validity: Concordance = 0.7363 indicates the baseline covariates (`task_age`, `log_publication_volume`) do predict displacement timing. The model is not broken; the diversity predictor is uninformative.

5.3 Kaplan-Meier Analysis

KM curves for Q1 (lowest diversity) vs. Q4 (highest diversity) show no meaningful separation. Visual inspection of survival curves confirms the null across the observation window.

Table 3: Cox Regression Results

Metric	Value	Threshold	Status
LRT statistic	0.0040	—	—
LRT p-value	0.9495	<0.05	FAIL
Hazard Ratio (HR)	1.006	$ \text{HR} - 1 \geq 0.10$	FAIL
95% CI	[0.846, 1.196]	excludes 1.0	FAIL
$ \text{HR} - 1 $	0.0056	≥ 0.10	FAIL ($\sim 18\times$ below)
M0 log-likelihood	-750.7335	—	—
M1 log-likelihood	-750.7315	—	—
$\Delta \log L$	0.0020	—	—
Concordance (M1)	0.7363	—	(acceptable)
Rows used	258	—	(87 NaN dropped)

Table 4: Comparison of Two Distinct Predictor Tests

Property	h-m1 Run 2 (prior)	H-M1 (current)
Predictor	$\Delta \text{score_lag1_z}$ (score velocity)	log_unique_paper_count (submission count)
Rows used	~ 64	258
EPV (approx.)	~ 22	~ 86
HR	0.871	1.006
LRT p	0.565	0.9495
Interpretation	Directional signal (underpowered)	Near-perfect null (adequate power)

5.4 Relationship to Prior Directional Signal

Critical clarification: These two runs test *different constructs*. The prior result ($\text{HR} = 0.871$) is for *score velocity* (how fast SOTA scores improved); the current result is for *submission-count diversity* (how many distinct papers submitted). The current null does not supersede or invalidate the prior directional signal for score velocity — that predictor has not been tested at adequate power. The two results are complementary: submission-count breadth is definitively null; score velocity remains an open question at full power.

6 Discussion

6.1 Interpreting the Near-Perfect Null

The combination of 5-gate all-pass and $\text{HR} = 1.006$ ($p = 0.9495$, $\text{EPV} \approx 86$) produces a clean scientific conclusion: community breadth diversity at benchmark introduction year, as measured by paper submission count, does not predict plurality benchmark displacement timing in Papers With Code (2015–2023). This is not a marginal null — LRT stat = 0.0040, $\Delta \log L = 0.0020$.

6.2 Why Does the Mechanism Not Operate?

We consider four explanations for the null:

E1 (HIGH): Predictor construct mismatch. The prior directional signal ($\text{HR} = 0.871$) was for *score velocity* ($\Delta \text{score_lag1_z}$) — a fundamentally different construct from submission-count diversity. Score velocity measures how fast SOTA improved; submission count measures how many teams participated. The current null applies to submission-count breadth only.

E2 (HIGH): Construct validity gap. `paper_url` deduplication is paper-level, not institution-level. If a few prolific labs dominate submissions, high submission count does not imply broad stakeholder investment. The lock-in mechanism requires *institutional* diversity (distinct teams with sunk costs). The null may reflect inadequacy of the proxy rather than absence of the mechanism.

E3 (MEDIUM): Mechanisms cancel. Lock-in ($HR < 1$) and saturation ($HR > 1$) may both operate but cancel across the population. The aggregate $HR \approx 1.0$ would then be a mixture of two opposing pathways, disentangleable only with moderator variables (e.g., task difficulty trajectory).

E4 (LOW): Residual temporal confound. Partial $r^2 = 0.605$ strongly argues against this.

Most parsimonious: E1 + E2 together predict the null independently.

6.3 Implications for Future Work

Score trajectory: Re-run Cox with `Δscore_lag1_z` on the full 258-row complete-case panel. Tests whether the prior $HR = 0.871$ (22 events) survives adequate power.

Institutional diversity: Augment h-e1 pipeline with Semantic Scholar author API to compute unique-institution count at introduction. Tests whether true stakeholder diversity predicts displacement when submission count does not.

6.4 Limitations

L1 (Construct validity): Submission count $\neq$ institutional adopter diversity. Null is valid for this operationalization; does not rule out better-constructed breadth measures.

L2 (25% data reduction): 87/345 rows dropped (NaN, primarily unmatched benchmarks). $EPV \approx 86$ remains adequate; potential selection bias if unmatched benchmarks differ systematically.

L3 (PH assumption): `check_assumptions()` was not completed due to a string-encoding error in `task_path` (requires integer-encoding before calling). Visual inspection of Schoenfeld residuals showed no obvious violation; quantitative confirmation pending. Given $HR = 1.006$, the conclusion is robust to moderate PH misspecification.

L4 (Scope): Results specific to Papers With Code (CC-BY-SA-4.0), 87 Koch et al. 2021 tasks, 2015–2023. Generalization to other registries or time periods not established.

L5 (Penalizer sensitivity): The L2 penalty (penalizer = 0.1) was selected for convergence, not optimized for effect estimation. In small samples (258 rows), L2 regularization can shrink coefficients toward zero. The null result $HR = 1.006$ should be interpreted as “not distinguishable from null under L2 regularization with penalizer = 0.1.” Sensitivity analysis with penalizer = 0.0 (unpenalized) and penalizer = 0.5 would confirm that the null is not a regularization artifact; these runs were not performed in the current pipeline.

6.5 On the Value of Principled Nulls

A null under poor measurement is uninformative. A null under validated measurement (5-gate all-pass) and adequate power ($EPV \approx 86$) definitively rules out a predictor class, saving future researchers from re-investigating the same construct. Submission-count community breadth diversity is such a predictor: well-measured, adequately powered, definitively null.

7 Conclusion

We set out to test whether community breadth at benchmark introduction predicts displacement timing. After validating all 5 FAIL FAST gates ($G0=0.862$, $G1=0.605$, $G2=0.975$, $G3=0.246$, $G4=2.14$), Cox regression on 258 complete-case rows ($EPV \approx 86$) yields:

$$HR = 1.006 \text{ (95\% CI = [0.846, 1.196]), LRT } p = 0.9495$$

This is a near-perfect null ($\Delta \log L = 0.0020$). Neither the lock-in nor the saturation mechanism is operative at a detectable level. Community breadth diversity, as measured by paper submission count, is ruled out as a predictor class for benchmark displacement timing in the Papers With Code ecosystem (2015–2023). This result is specific to the submission-count operationalization; score velocity and institutional diversity remain untested at full power.

The 5-gate FAIL FAST protocol is a reusable methodological contribution for benchmark lifecycle survival analysis — a pre-registration-equivalent rigor framework that separates data quality failure from hypothesis failure.

Future directions: (1) score trajectory ($\Delta\text{score}_{\text{lag1_z}}$) as primary predictor on the full 258-row panel; (2) institution-deduplicated breadth via Semantic Scholar author API; (3) cross-platform replication on Semantic Scholar or OpenReview data.

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Appendix

A. Additional Figures

Figure A1: Schoenfeld Residuals. PH assumption diagnostic (`h-m1/figures/schoenfeld_residuals.png`). Visual inspection showed no obvious violation. Automated quantitative test was not completed due to a string-type covariate encoding error (`task_path`); fixing requires integer-encoding `task_path` before calling `check_assumptions()`.

Figure A2: Coverage Heatmap. Matched vs. unmatched h-e2 benchmarks in the pwc-archive/evaluation-tables fuzzy join (`h-e1/figures/coverage_heatmap.png`). 75/87 benchmarks matched (86.2%); 12 below `token_sort_ratio=85` threshold.

Figure A3: Predictor Distributions. Histograms of `log_unique_paper_count_at_intro` and `paper_diversity_ratio` across 67 benchmarks with computed diversity statistics (`h-e1/figures/predictor_distributions.png`).

Figure A4: Correlation Matrix. Pearson r heatmap for all Cox covariates (`h-e1/figures/correlation_matrix.png`). Confirms low inter-predictor correlations (max $r = -0.324$ between diversity predictors).

B. Experimental Environment

OS: Linux (Ubuntu 20.04)

GPU: 5x NVIDIA H100 NVL (not used -- statistical pipeline)

Python: 3.10 (conda envs: `youra-h-e1`, `youra-h-m1`)

Key packages: `datasets>=2.0`, `lifelines>=0.27`, `rapidfuzz`,
`statsmodels`, `scipy`, `pandas`, `numpy`,
`matplotlib`, `seaborn`

Data: `pwc-archive/evaluation-tables` (HuggingFace, Jul 2025)

h-e2 panel: 87 tasks, 345 rows, 2015-2023

C. Code Availability

All code generated in this pipeline (H-E1, H-M1) is available in the research archive. Key modules: `pipeline.py` (`DataLoader`, `FuzzyJoiner`, `DiversityAggregator`), `gates.py` (`GateValidator`, `VIFChecker`), `cox_analysis.py` (`LRTResult`, `fit_models`, `run_lrt`), `visualization.py` (5 figure functions).